# 50Days-coding-challenge

## Day-14[SQL]

## SQL Coding Challenge: Sales Analysis

**Name:M NITHIN**

### Table: Sales

CREATE TABLE Sales ( sale_id INT PRIMARY KEY, customer_name VARCHAR(50), city VARCHAR(50), product_name VARCHAR(50), category VARCHAR(50), quantity INT, price DECIMAL(10,2), sale_date DATE );

### Sample Data

INSERT INTO Sales VALUES (1, 'Ravi', 'Bangalore', 'Laptop', 'Electronics', 1, 60000, '2023-06-01'), (2, 'Priya', 'Chennai', 'Mobile', 'Electronics', 2, 20000, '2023-06-03'), (3, 'Amit', 'Delhi', 'Headphones', 'Accessories', 3, 2000, '2023-06-05'), (4, 'Neha', 'Mumbai', 'Chair', 'Furniture', 2, 5000, '2023-06-07'), (5, 'Ravi', 'Bangalore', 'Mobile', 'Electronics', 1, 20000, '2023-06-10'), (6, 'Priya', 'Chennai', 'Laptop', 'Electronics', 1, 60000, '2023-06-12'), (7, 'Amit', 'Delhi', 'Chair', 'Furniture', 4, 5000, '2023-06-15'), (8, 'Neha', 'Mumbai', 'Headphones', 'Accessories', 2, 2000, '2023-06-18');

## Challenge Questions

1. Show all records from the Sales table.

Query  Query History

```
(3, 'Amit', 'Delhi', 'Headphones', 'Accessories', 3, 2000, '2023-06-05'),
(4, 'Neha', 'Mumbai', 'Chair', 'Furniture', 2, 5000, '2023-06-07'),
(5, 'Ravi', 'Bangalore', 'Mobile', 'Electronics', 1, 20000, '2023-06-10'),
(6, 'Priya', 'Chennai', 'Laptop', 'Electronics', 1, 60000, '2023-06-12'),
(7, 'Amit', 'Delhi', 'Chair', 'Furniture', 4, 5000, '2023-06-15'),
(8, 'Neha', 'Mumbai', 'Headphones', 'Accessories', 2, 2000, '2023-06-18');

Select*from sales;
```

Data Output  Messages  Notifications

Showing rows: 1 to 8  Page No: 1 of 1

| | sale_id [PK] integer | customer_name character varying (50) | city character varying (50) | product_name character varying (50) | category character varying (50) | quantity integer | price numeric (10,2) | sale_date date |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | Ravi | Bangalore | Laptop | Electronics | 1 | 60000.00 | 2023-06-01 |
| 2 | 2 | Priya | Chennai | Mobile | Electronics | 2 | 20000.00 | 2023-06-03 |
| 3 | 3 | Amit | Delhi | Headphones | Accessories | 3 | 2000.00 | 2023-06-05 |
| 4 | 4 | Neha | Mumbai | Chair | Furniture | 2 | 5000.00 | 2023-06-07 |
| 5 | 5 | Ravi | Bangalore | Mobile | Electronics | 1 | 20000.00 | 2023-06-10 |
| 6 | 6 | Priya | Chennai | Laptop | Electronics | 1 | 60000.00 | 2023-06-12 |
| 7 | 7 | Amit | Delhi | Chair | Furniture | 4 | 5000.00 | 2023-06-15 |
| 8 | 8 | Neha | Mumbai | Headphones | Accessories | 2 | 2000.00 | 2023-06-18 |

2. Display only customer names and cities.

Sales Analysis/postgres@PostgreSQL 17

No limit

Query | Query History

```
(6, 'Priya', 'Chennai', 'Laptop', 'Electronics', 1, 60000, '2023-06-12'),
(7, 'Amit', 'Delhi', 'Chair', 'Furniture', 4, 5000, '2023-06-15'),
(8, 'Neha', 'Mumbai', 'Headphones', 'Accessories', 2, 2000, '2023-06-18');

Select*from sales;

select customer_name,city
from sales;
```

Data Output | Messages | Notifications

Showing rows: 1 to 8 | Page No: 1 of 1

| | customer_name character varying (50) | city character varying (50) |
|---|---|---|
| 1 | Ravi | Bangalore |
| 2 | Priya | Chennai |
| 3 | Amit | Delhi |
| 4 | Neha | Mumbai |
| 5 | Ravi | Bangalore |
| 6 | Priya | Chennai |
| 7 | Amit | Delhi |
| 8 | Neha | Mumbai |

3. Show product name and price.

Sales Analysis/postgres@PostgreSQL 17

No limit

Query | Query History

```
Select*from sales;

select customer_name,city
from sales;

select product_name,price
from sales;
```

Data Output | Messages | Notifications

Showing rows: 1 to 8 | Page No: 1 of 1

| | product_name character varying (50) | price numeric (10,2) |
|---|---|---|
| 1 | Laptop | 60000.00 |
| 2 | Mobile | 20000.00 |
| 3 | Headphones | 2000.00 |
| 4 | Chair | 5000.00 |
| 5 | Mobile | 20000.00 |
| 6 | Laptop | 60000.00 |
| 7 | Chair | 5000.00 |
| 8 | Headphones | 2000.00 |

Total rows: 8 | Query complete 00:00:00.116 | CRLF | Ln 25, Col 12

4. Show all sales where city is 'Bangalore'.

Sales Analysis/postgres@PostgreSQL 17

No limit

Query Query History

```
select customer_name,city
from sales;

select product_name,price
from sales;

select*from sales
where City='Bangalore';
```

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

| | sale_id [PK] integer | customer_name character varying (50) | city character varying (50) | product_name character varying (50) | category character varying (50) | quantity integer | price numeric (10,2) | sale_date date |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | Ravi | Bangalore | Laptop | Electronics | 1 | 60000.00 | 2023-06-01 |
| 2 | 5 | Ravi | Bangalore | Mobile | Electronics | 1 | 20000.00 | 2023-06-10 |

Total rows: 2 Query complete 00:00:00.157 CRLF Ln 28, Col 24

5. Find all products in 'Electronics' category.

Sales Analysis/postgres@PostgreSQL 17

No limit

Query Query History

```
select product_name,price
from sales;

select*from sales
where City='Bangalore';

select*from sales
where category='Electronics';
```

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1 of 1

| | sale_id [PK] integer | customer_name character varying (50) | city character varying (50) | product_name character varying (50) | category character varying (50) | quantity integer | price numeric (10,2) | sale_date date |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | Ravi | Bangalore | Laptop | Electronics | 1 | 60000.00 | 2023-06-01 |
| 2 | 2 | Priya | Chennai | Mobile | Electronics | 2 | 20000.00 | 2023-06-03 |
| 3 | 5 | Ravi | Bangalore | Mobile | Electronics | 1 | 20000.00 | 2023-06-10 |
| 4 | 6 | Priya | Chennai | Laptop | Electronics | 1 | 60000.00 | 2023-06-12 |

Total rows: 4 Query complete 00:00:00.110 CRLF Ln 30, Col 1

6. Show sales where price is greater than 10,000.

Sales Analysis/postgres@PostgreSQL 17

No limit

Query | Query History

```
select*from sales
where City='Bangalore';

select*from sales
where category='Electronics';

select*from sales
where price>10000;
```

Data Output | Messages | Notifications

Showing rows: 1 to 4 Page No: 1 of 1

| | sale_id [PK] integer | customer_name character varying (50) | city character varying (50) | product_name character varying (50) | category character varying (50) | quantity integer | price numeric (10,2) | sale_date date |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | Ravi | Bangalore | Laptop | Electronics | 1 | 60000.00 | 2023-06-01 |
| 2 | 2 | Priya | Chennai | Mobile | Electronics | 2 | 20000.00 | 2023-06-03 |
| 3 | 5 | Ravi | Bangalore | Mobile | Electronics | 1 | 20000.00 | 2023-06-10 |
| 4 | 6 | Priya | Chennai | Laptop | Electronics | 1 | 60000.00 | 2023-06-12 |

Total rows: 4 Query complete 00:00:00.117 CRLF Ln 33, Col 1

7. Display sales where quantity is more than 2.

Sales Analysis/postgres@PostgreSQL 17

No limit

Query | Query History

```
select*from sales
where category='Electronics';

select*from sales
where price>10000;

select*from sales
where quantity>2;
```

Data Output | Messages | Notifications

Showing rows: 1 to 2 Page No: 1 of 1

| | sale_id [PK] integer | customer_name character varying (50) | city character varying (50) | product_name character varying (50) | category character varying (50) | quantity integer | price numeric (10,2) | sale_date date |
|---|---|---|---|---|---|---|---|---|
| 1 | 3 | Amit | Delhi | Headphones | Accessories | 3 | 2000.00 | 2023-06-05 |
| 2 | 7 | Amit | Delhi | Chair | Furniture | 4 | 5000.00 | 2023-06-15 |

Total rows: 2 Query complete 00:00:00.120 CRLF Ln 37, Col 18

8. Show all sales sorted by price (low to high).

Sales Analysis/postgres@PostgreSQL 17

No limit

Query | Query History

```
select*from sales
where price>10000;

select*from sales
where quantity>2

select*from sales
order by price asc;
```

Data Output | Messages | Notifications

Showing rows: 1 to 8 Page No: 1 of 1

| | sale_id [PK] integer | customer_name character varying (50) | city character varying (50) | product_name character varying (50) | category character varying (50) | quantity integer | price numeric (10,2) | sale_date date |
|---|---|---|---|---|---|---|---|---|
| 1 | 8 | Neha | Mumbai | Headphones | Accessories | 2 | 2000.00 | 2023-06-18 |
| 2 | 3 | Amit | Delhi | Headphones | Accessories | 3 | 2000.00 | 2023-06-05 |
| 3 | 7 | Amit | Delhi | Chair | Furniture | 4 | 5000.00 | 2023-06-15 |
| 4 | 4 | Neha | Mumbai | Chair | Furniture | 2 | 5000.00 | 2023-06-07 |
| 5 | 5 | Ravi | Bangalore | Mobile | Electronics | 1 | 20000.00 | 2023-06-10 |
| 6 | 2 | Priya | Chennai | Mobile | Electronics | 2 | 20000.00 | 2023-06-03 |
| 7 | 6 | Priya | Chennai | Laptop | Electronics | 1 | 60000.00 | 2023-06-12 |
| 8 | 1 | Ravi | Bangalore | Laptop | Electronics | 1 | 60000.00 | 2023-06-01 |

Total rows: 8 Query complete 00:00:00.122 CRLF Ln 38, Col 1

9. Display sales sorted by quantity (high to low).

Sales Analysis/postgres@PostgreSQL 17

No limit

Query | Query History

```
select*from sales
where quantity>2

select*from sales
order by price asc;

select*from sales
order by quantity desc;
```

Data Output | Messages | Notifications

Showing rows: 1 to 8 Page No: 1 of 1

| | sale_id [PK] integer | customer_name character varying (50) | city character varying (50) | product_name character varying (50) | category character varying (50) | quantity integer | price numeric (10,2) | sale_date date |
|---|---|---|---|---|---|---|---|---|
| 1 | 7 | Amit | Delhi | Chair | Furniture | 4 | 5000.00 | 2023-06-15 |
| 2 | 3 | Amit | Delhi | Headphones | Accessories | 3 | 2000.00 | 2023-06-05 |
| 3 | 8 | Neha | Mumbai | Headphones | Accessories | 2 | 2000.00 | 2023-06-18 |
| 4 | 4 | Neha | Mumbai | Chair | Furniture | 2 | 5000.00 | 2023-06-07 |
| 5 | 2 | Priya | Chennai | Mobile | Electronics | 2 | 20000.00 | 2023-06-03 |
| 6 | 5 | Ravi | Bangalore | Mobile | Electronics | 1 | 20000.00 | 2023-06-10 |
| 7 | 1 | Ravi | Bangalore | Laptop | Electronics | 1 | 60000.00 | 2023-06-01 |
| 8 | 6 | Priya | Chennai | Laptop | Electronics | 1 | 60000.00 | 2023-06-12 |

Total rows: 8 Query complete 00:00:00.141 CRLF Ln 42, Col 1

10. Show unique cities.

Sales Analysis/postgres@PostgreSQL 17

No limit

Query | Query History

```
38
39  select*from sales
40  order by price asc;
41
42  select*from sales
43  order by quantity desc;
44
45  select distinct city from sales;
```

Data Output | Messages | Notifications

Showing rows: 1 to 4 | Page No: 1 of 1

| | city<br>character varying (50) |
|---|---|
| 1 | Mumbai |
| 2 | Delhi |
| 3 | Chennai |
| 4 | Bangalore |

Total rows: 4 | Query complete 00:00:00.115 | CRLF | Ln 45, Col 1